

Description of keywords of VTD (VLBI Time Delay) control file language

L. Petrov

Abstract:

This document provides detailed description of syntax of the language used for specification of control file for the package VTD (VLBI Time Delay).

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1 General rules

Control file for VTD consists of lines of variable length. Lines which starts from characters # or * are considered as comments and ignored by parsing software. Each line consists of a keyword and the value. All keywords must be specified, no defaults are allowed. A value should be separated the keyword by one or more delimiters. Valid delimiters: blank (decimal code 32), tabulation (decimal code 9) and binary zero. The first line of the control file should have the label of the format version. The current label version is
VTD Control file. Format version of 2009.02.21

If the file does not have correct format label, VTD library will issue and errors message.

2 Description of keyword

2.1 LEAP_SECOND:

LEAP_SECOND: [leap_second_file_name or NERS]

leap_second_file_name is the name of the file in LEAP_SECOND format which tabulates epochs of discontinuities of the function $UT1 - TAI(t)$ and values of discontinuities.

If NERS is specified, the leap second is downloaded automatically using the Network Earth Rotation Service

2.2 DE_EPHEMERIDES:

DE_PHEMERIDES: planetary_ephemerides_file_name

Planetary_ephemerides_file_name is the name of the DE ephemerides file.

2.3 STATION_DESCRIPTION:

STATION_DESCRIPTION: station_description_file_name

Station_description_file_name is the name of the file in STADESC format. That file contains information about all stations which participated in the experiment: IVS station name, antenna mounting type, antenna axis offset, tectonic plate. This file may contain information about other stations which did not participated in the analyzed experiment. All stations which participated in the experiment must be listed in this file.

2.4 STATION_COORDINATES:

STATION_COORDINATES: station_coordinates_file_name

Station_coordinates_file_name is the name of the file either in GETPAR_STA format
(http://astrogeo.org/psolve/doc/getpar_02.html#section3.1)
or in SIT-MODFILE format. That file contains information about all stations which participated in the experiment: IVS station name, station coordinates and epoch for coordinates of those stations which had episodic motion. This file may contain information about other stations which did not participated in the analyzed experiment. All stations which participated in the experiment must be listed in this file.

2.5 STATION_VELOCITIES:

STATION_VELOCITIES: station_velocities_file_name

Station_velocities_file_name is the name of the file either in GETPAR_VEL format
(http://astrogeo.org/psolve/doc/getpar_02.html#section3.2)
or in VEL-MODFILE format. That file contains information about all stations which participated in the experiment: IVS station name, station velocities. This file may contain information about other stations which did not participated in the analyzed experiment. All stations which participated in the experiment must be listed in this file.

2.6 STATION_ECCENTRICITIES:

STATION_ECCENTRICITIES: station_eccentricities_file_name

Station_eccentricities_file_name is the name of the file with eccentricity vector in ECC-FORMAT. It contains IVS station name, CDP monument number, starting date of validity of the eccentricity vector, end date of validity of the eccentricity vector, eccentricity vector for each station each period of validity defined as the vector from the monument point for which station coordinates are referred to, to the reference point on the antenna, defined as a point on the fixed axis which is located at the closest distance to the moving axis.

2.7 SOURCE_COORDINATES:

SOURCE_COORDINATES: source_coordinates_file_name

The qualifier Source_coordinates_file_name is the name of the file either in GETPAR_SOU format

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(http://astrogeo.org/psolve/doc/getpar_02.html#section3.3)
or in SOU-MODFILE format. That file contains information the sources which participated in the experiment: IVS source name, source coordinates. This file may contain information about other source which were not observed in the analyzed experiment.

2.8 SOURCE_COORDINATES_2ND:

SOURCE_COORDINATES: [source_coordinates_file_name_2 or NONE]

The qualifier source_coordinates_file_name is the name of the file either in GETPAR_SOU format
(http://astrogeo.org/solve_root/help/getpar_02.html#section3.3)
or in SOU-MODFILE format. This is the secondary catalogue used by VTD. It might be convenient to keep coordinates of source in more than one file. That file contains information the sources which participated in the experiment: IVS source name, source coordinates. This file may contain information about other source which were not observed in the analyzed experiment.

2.9 SOURCE_COORDINATES_3RD:

SOURCE_COORDINATES: [source_coordinates_file_name_3 or NONE]

The qualifier source_coordinates_file_name is the name of the file either in GETPAR_SOU format
(http://astrogeo.org/psolve/doc/getpar_02.html#section3.3)
or in SOU-MODFILE format. This is the ternary catalogue used by VTD. It might be convenient to keep coordinates of source in more than one file. That file contains information the sources which participated in the experiment: IVS source name, source coordinates. This file may contain information about other source which were not observed in the analyzed experiment.

2.10 SOURCE_COORDINATES_4TH:

SOURCE_COORDINATES: [source_coordinates_file_name_4 or NONE]

The qualifier source_coordinates_file_name is the name of the file either in GETPAR_SOU format
(http://astrogeo.org/psolve/doc/getpar_02.html#section3.3)
or in SOU-MODFILE format. This is the fourth catalogue used by VTD.
It might be convenient to keep coordinates of source in more than one file.
That file contains information the sources which participated in the experiment: IVS source name, source coordinates. This file may contain information about other source which were not observed in the analyzed experiment.

2.11 SOURCE_PRLX_PRP_MOTION:

SOURCE_PRLX_PRP_MOTION: [source_parallax_and_proper_motion_file or NONE]

The qualifier source_parallax_and_proper_motion_file specifies the name of the file with a priori source proper motions and parallaxes. Each line of that file contains IVS source name J2000 source name, proper motion over right ascension without scaling $\cos(\delta)$ factor, proper motion over declination, annual parallax, and comment. See file prlx_prp_format.txt in the documentation directory for the format description.

2.12 DIR_BTXX:

DIR_BTXX: [directory_name or NONE]

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Directory with files with phase center offsets in bintex format for both space-born and ground based antennas. Files in bintex format are created by utility vtd_antex_to_bintex that processes an input file in the IGS ascii format ANTEX into output files. VTD searches for phase center offsets in the directory tree specified by DIR_BTX keyword.

2.13 DIR_NZO:

DIR_NZO: [directory_name or NONE]

Directory with files with orbits of near-zone objects. Supported formats for orbit files are:

NZO -- Internal VTD ascii format that tabulates time series of positions and velocities.

ORM -- The Consultative Committee for Space Data Systems orbit messages, CCSDS 502.0-B-3, <https://ccsds.org/Pubs/502x0b3e1.pdf>

TLE -- the two line element format, <https://www.celertrak.org/columns/v04n03/>

SP3 -- The Extended Standard Product 3 Orbit Format (SP3-c) <https://files.igs.org/pub/data/format/sp3c.txt>

2.14 AEM_FILE

AEM_FILE: aem_file_name or NONE

Keyword AEM_FILE specifies the name of the file with the a priori Earth rotation model. This file is one of the three files that describes the Earth's rotation in accordance to the empirical Earth rotation model formalism. The file should conform specifications of the AEM files. The AEM file defines values of "large" parameters, i.e. those parameters which squares are not-negligible. These are parameters for precession, several terms in nutation and several terms in the Earth's axial rotation.

2.15 ERM_FILE

ERM_FILE: erm_file_name or NONE

Keyword ERM_FILE specifies the name of the file with the a priori Earth rotation model. This file is one of the three files that describes the Earth's rotation in accordance to the empirical Earth rotation model formalism. The file should conform specifications of the ERM files. The ERM file defines coefficients that describe long periodic, "small" variations in Earth's rotation, i.e. those parameters variations which squares are negligible. The coefficients of B-spline allows to compute the instantaneous vector of perturbations to the Earth's rotation. These file describes the slowly varying component of the Earth rotation.

2.16 HARMONIC_EOP_FILE:

HARMONIC_EOP_FILE: [NONE or NERS or file_name]

Keyword HARMONIC_EOP_FILE specifies the name of the file with the harmonic Earth Orientation model. This models describes perturbation in the Earth orientation with respect to the a priori model specified by the a priori EOP files, a priori nutation, precession and built-in expression for the UT1 angle. The model is in the form

$$\sum (a + b*t)*\{\sin/\cos\}(\theta/2*t^2 + \omega*t + \text{phase})$$

The purpose of this model is to correct a priori EOP model specified in EOP_SERIES, PRECESSION_EXPRESSION, NUTATION_EXPANSION, GEODESIC_NUTATION keywords.

NERS -- the harmonic EOP variations are taken using the Network Earth Rotation Service (NERS) automatically. The NERS client control file specified by EOP_SERIES keyword is used.

file_name -- name of the file with the model in HEO format.

2.17 EOP_SERIES:

EOP_SERIES: [NONE or NERS or ners_configuration_file or earth_orientation_parameters_file]

Earth_orientation_parameters_file is the name of the file of the Earth orientation parameters either in EOP-MOD or NERS EOP series format, or NERS configuration file or NERS. The file with the Earth Orientation Parameters contains values of X-coordinate of the pole coordinate, Y-coordinate of the pole coordinate, and UT1 angle at the equidistant grid of time epochs.

NONE -- No external file with EOP series.

(Supported when ERM formalism was chosen)

NERS -- The EOP data are taken using NERS. Default NERS client control file is used.

ners_configuration_file -- name of the control file for NERS client

earth_orientation_parameters_file -- name of the file with Earth orientation parameters

VTD analyzes the first line of the file in order to determine whether it is the file with the Earth orientation parameters or the NERS configuration file.

2.18 EOP_TIME_SCALE:

EOP_TIME_SCALE: time_scale_of_eop_table

Time_scale_of_eop_table defines the time scale used for time tags in the table of the input Earth orientation series specified in the value of the keyword EOP_SERIES. Supported values:

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TAI
TDT
TDB
UTC
UT1
NERS -- the same as TAI
NONE -- (Supported when ERM formalism was chosen)

2.19 UZT_MODEL:

UZT_MODEL: [NONE or NERS or DICKMAN1993 or DICKMAN1993_PRINCIPAL or DICKMAN1993_SHORT]

UT1_zonal_tides_model_name is the name of the model of UT1 variations induced by zonal tides. Supported values:

NONE -- no model is used.
NERS -- the same as NONE
DICKMAN1993 -- model of Dickman, 1993, all terms
DICKMAN1993_PRINCIPAL -- model of Dickman, 1993, only principal term and 14 terms which contributes to UT1 rate greater than 1.0D-14 rad/s
DICKMAN1993_SHORT -- model of Dickman, 1993, only terms with periods shorter than 60 days

2.20 UZT_USE:

UZT_USE: [NONE or NERS or ADD or INTERPOLATE or SUBTRACT]

The way how to apply the mode of UT1 variations induced by zonal tides.

NONE -- not to apply contribution of UT1 variations induced by zonal tides;
NERS -- the same as NONE

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ADD -- add contribution to UT1 caused by zonal tides to apriori EOP;

SUBTRACT -- subtract contribution to UT1 caused by zonal tides from a priori EOP;

INTERPOLATE -- the model is used for subtracting contribution to UT1 due to zonal tides before computation of interpolating coefficients, and then contribution of UT1 are added to final a priori UT1 values after interpolation.

2.21 EROT_COMPAT:

EROT_COMPAT: [NONE or NERS or CALC10 or NO_SANG]

If specified, adds corrections to the a priori Earth rotation angles to be compatible with Calc10. It can be used for testing in order to precisely emulate values computed by Calc. These corrections DO NOT improve VLBI path delay prediction. This option is intended for tests only.

NONE (recommended) means that no emulation is performed.

NERS the same as NONE

2.22 PRECESSION_EXPRESSION:

PRECESSION_EXPRESSION: precession_model_name

Keyword PRECESSION_EXPRESSION specifies the name of the precession model. Supported expansions:

LIESE_1977 -- Values of precession parameters from Lieske 1976.

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SIMON_1994 -- Values of precession parameters from Simon 1994.
IERS_1996 -- Values of precession parameters from IERS 1996.
CAPITAINE_2003 -- Values of precession parameters from Capitaine 2003
CAPITAINE_2005 -- Values of precession parameters from Capitaine 2005
NERS -- Precession is computed by the NERS. The NERS client
control file specified by EOP_SERIES keyword is used.
NONE -- No precession (Supported when ERM formalism was chosen)

2.23 NUTATION_EXPANSION:

NUTATION_EXPANSION: nutation_expansion_name

Keyword NUTATION_EXPANSION specifies the name of the nutation expansions.
Supported expansions:

WAHR1980 -- Wahr 1980 nutation expansion. Expansion for Doodson
arguments referred to at that paper are used.
IERS1996 -- Empirical nutation expansion computed by T. Herring
and presented in the IERS 1996 Conventions. Expansion
for Doodson arguments referred to at that paper are used.
REN2000 -- Theoretical expansion of the rigid Earth model by
J. Souchay
MHB2000 -- MHB2000 expansion according coefficients presented in
the program which T. Herring posted on the Web.
MHB2000_TRANSF -- REN2000 expansion with the transfer function from
Mathews et al, 2002 paper.
MHB2000_ADDON -- REN2000 expansion with the transfer function from
Mathews et al, 2002 paper and all add-on terms listed
at that paper.
MHB2000_PETA -- Simplified expansion to nutation with only three non-linear
terms.
NERS -- Nutation is computed by the NERS. The NERS client
control file specified by EOP_SERIES keyword is used.
NONE -- No nutation (Supported when ERM formalism was chosen)

2.24 GEODESIC_NUTATION:

GEODESIC_NUTATION usage_flag

Keyword NUTATION_EXPANSION specifies whether geodesic nutation should be modeled.

usage_flag -- one of
 YES -- to model geodesic nutation
 NONE -- not to model geodesic nutation

2.25 SOLID_EARTH_TIDES_ZERO_FREQ:

SOLID_EARTH_TIDES_ZERO_FREQ: Love_model_name

Keyword SOLID_EARTH_TIDES_ZERO_FREQ specifies the name of the model of Love numbers for the zeroth frequency. Unfortunately, several erroneous papers and thoughtless resolution of scientific societies created a great confusion about the issues of so-called "permanent tide". VTD allows a user to specify the model for computation of the zonal tides of the zero-th frequency.

Love_model_name -- model code. One of

 MDG97AN -- Mathews, Dehant, Gipson 1997, anelastic
 variant.
 NONE -- consider Love numbers for the tides of
 the second degree as zero.

2.26 SOLID_EARTH_TIDES_2ND_DEGREE:

SOLID_EARTH_TIDES_2ND_DEGREE: Love_model_name

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Keyword `SOLID_EARTH_TIDES_2ND_DEGREE` specifies the name of the model of Love numbers of the second degree as a function of frequency.

`Love_model_name` -- model code. One of

`MDG97EL` -- Mathews, Dehant, Gipson 1997, elastic variant.
`MDG97AN` -- Mathews, Dehant, Gipson 1997, anelastic variant.
`DDW99EH` -- Dehant, Defraigne, Wahr 1999, elastic, hydrostatic variant.
`DDW99IN` -- Dehant, Defraigne, Wahr 1999, inelastic, non-hydrostatic variant.
`LOVE` -- Love, 1923 model
`MATHEWS_2000` -- Mathews, 2000 model
`MATHEWS_2001` -- Mathews, 2001 model
`NONE` -- consider Love numbers for the tides of the second degree as zero.

2.27 `SOLID_EARTH_TIDES_3RD_DEGREE`:

`SOLID_EARTH_TIDES_3RD_DEGREE: Love_model_name`

Keyword `SOLID_EARTH_TIDES_2ND_DEGREE` specifies the name of the model of Love numbers of the third degree.

`Love_model_name` -- model code. One of

`MDG97EL` -- Mathews, Dehant, Gipson 1997.
`NONE` -- consider Love numbers for the tides of the second degree as zero.

2.28 `POLE_TIDE_MODEL`:

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POLE_TIDE_MODEL: Love_model_name

Keyword POLE_TIDE_MODEL specifies the name of the model of Love numbers used for computation of the pole tide.

Love_model_name -- model code. One of

MDG97EL -- Mathews, Dehant, Gipson 1997, elastic
variant.
MDG97AN -- Mathews, Dehant, Gipson 1997, anelastic
variant.
DDW99EH -- Dehant, Defraigne, Wahr 1999, elastic,
hydrostatic variant.
DDW99IN -- Dehant, Defraigne, Wahr 1999, inelastic,
non-hydrostatic variant.
LOVE -- Love, 1923 model
MATHEWS_2000 -- Mathews, 2000 model
MATHEWS_2001 -- Mathews, 2001 model
NONE -- consider Love numbers for the pole tides
as zero.

2.29 MPL_FILE:

MEAN_POLE_MODEL: expression_id

Keyword MEAN_POLE_MODEL MPL_FILE specifies the name of the expression for the mean pole positions.

expression_id -- one of

NONE -- mean pole positions is zero.

IERS2010 -- expression for pole tide following IERS2010 conventions.
NB: THIS EXPRESSION IS WRONG!! It was included only
for tests.

IERS2022 -- expression for pole tide following IERS2022 conventions.

2.30 POSVAR_FIL:

POSVAR_FIL: model_index model_file_name

Keyword POSVAR_FIL specifies the name of the file with the model of position variations.

model_index -- the index of the model. Should be 1 if only one model of position variations is in use. Of more than one model is in use, then the models are numbered consecutively from 1 to the last model. Maximum 8 models can be specified.

model_file_name -- the name of the file with the model.

2.31 POSVAR_MOD:

POSVAR_MOD: model_index model_type

Keyword POSVAR_MOD specifies the model type. Two types are supported: time series and harmonic site position variations model.

model_index -- the index of the model. Should be 1 if only one model of position variations is in use. Of more than one model is in use, then the models are numbered consecutively from 1 to the last model. Maximum 8 models can be specified.

model_type -- one of
HARMONIC_MODE -- the file is in HARPOS format:
http://astrogeo.org/psolve/doc/harpos_format.txt
TIME_SERIES -- the file is in BINDISP format:
http://astrogeo.org/psolve/doc/bindisp_format.txt
B_SPLINE -- ...

2.32 POSVAR_INT:

POSVAR_INT: model_index interpolation_type

Keyword POSVAR_MOD specifies the type of interpolation. The values of the position variations models are first read at several epoch around the time range of the experiment, then interpolation polynomials are computed. Position variations at the epoch of the observation are computed on the basis of this polynomial. Two interpolation types are supported: LINEAR and SPLINE

```
model_index      -- the index of the model. Should be 1 if only one
                  model of position variations is in use. Of more than
                  one model is in use, then the models are numbered
                  consecutively from 1 to the last model. Maximum
                  8 models can be specified.
interpolation_type -- one of
                  LINEAR
                  SPLINE
```

2.33 POSVAR_USE:

POSVAR_USE: model_index usage_flag

Keyword POSVAR_INT specifies the action of position variations model is not available for the requested station, requested epochs. An alternative is either issue a warning and continue or print the fatal error message and stop.

```
model_index      -- the index of the model. Should be 1 if only one
                  model of position variations is in use. Of more than
                  one model is in use, then the models are numbered
                  consecutively from 1 to the last model. Maximum
                  8 models can be specified.
interpolation_type -- one of
                  REQUIRED  -- if position variation model is not
                           available for a requested station,
                           requested epoch, then stop.
```

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IF_AVAILABLE -- if position variation model is not available for a requested station, requested epoch, then print a warning, set variations to zero and continue.

2.34 AXIS_OFFSET_MODEL:

AXIS_OFFSET_MODEL: usage_flag

usage_flag -- the flag. Supported values:

YES -- (recommended) means that the contribution to delay due to antenna axis offset is taken into account.

NONE -- means that the contribution to delay caused by the antenna axis offset is ignored. It is used for tests only.

CALC9 -- forces VTD to emulate a bug in Calc9. This option is for tests only.

2.35 ANTENNA_DEFORMATIONS_FILE:

ANTENNA_DEFORMATIONS_FILE: [NONE or agd_file]

Keyword ANTENNA_DEFORMATIONS_FILE specifies the file name that keeps parameters of the models of antenna deformation caused by gravity. The file should conform AGD specifications. The current model tabulates the change of the focal length as a function of elevation for each antenna. Some antenna may be missing from the agd_file file. The contribution to delay caused by antenna deformations for these antenna will be set to zero.

2.36 ANTENNA_THERMAL_EXPANSION

**ANTENNA_THERMAL_EXPANSION NONE or [MODEL antenna_information_file
INSITU [NONE or insitu_file]
TEMPERATURE [METEO or SPD]
LAG time_in_seconds]**

Keyword ANTENNA_THERMAL_EXPANSION defines the mode for computing the contribution to time delay caused by antenna structures thermal expansion. The 1st qualifier may be NONE or MODEL. If the 1st qualifier is MODEL, then VTD expects to read 8 qualifiers. The second qualifier is the name of the antenna information file in the ANTI format. The antenna information file defines the type of each antenna, dimensions of the antenna structure and their coefficient of thermal expansion, reference temperature. Some antennas may be missing from this file. Contribution to path delay caused by antenna deformations of these antennas will be set to zero. The 3rd qualifier is INSITU. The 4th qualifier is the name of the file with information about antennas that have available data of their thermal height variation measurements with the invar rod. These stations will be excluded from computation of contribution due to antenna thermal deformation according to the model of thermal expansion. Position variations of the reference point for these antennas should be included to the model specified by POSVAR_MOD, POSVAR_FIL, POSVAR_INT, POSVAR_USE.

The antenna deformations caused by thermal expansion depend on the effective temperature of elements of antenna structure. The 5th qualifier is TEMPERATURE. It defines the source of temperature data defined in the 6th qualifier METEO or SPD. If METEO, then temperature is supplied with VTD_METEO_IN subroutine. These are usually data collected by local meteorological sensor. If SPD, then air temperature is supplied in SPD (slant path delay) data files. It is computed from the output of the global numerical weather model. The 7th qualifier is LAG. The 8th qualifier defines the lag of antenna temperature with respect to air temperature. Antenna temperature at time t is computed as air temperature at time $t - \text{lag}$.

Restrictions: TEMPERATURE SPD requires EXTERNAL_DELAY_DIR directory names be defined that supplied the source of information about expansion coefficients via 3D B-splines.

2.37 METEO_DEF

METEO_DEF model

Qualifier METEO_DEF defines the model that is used for computing missing meteorological parameters. If the atmospheric pressure and/or air temperature are negative, they are interpreted as missing.

default_model:

IMA -- the model of the International Meteorological Association
ISO Standard atmosphere 1976.

CALC -- Undocumented model that Calc uses. This model is for
test only.

NONE -- not to change meteorological parameters. Keep in mind,
that if NONE specified, subsequent computation of zenith
path delay using SAASTAMOINEN model will be
___completely wrong___.

2.38 HYDROSTATIC_ZENITH_DELAY:

**HYDROSTATIC_ZENITH_DELAY: [NONE or zenith_delay_path_model_name or
MMF mmd_model_file]**

Keyword HYDROSTATIC_ZENITH_DELAY specifies the name of the a priori model for the hydrostatic component zenith path delay.

zenith_delay_path_model_name -- the name of the model. One of

NONE -- do not model hydrostatic path delay

SAASTAMOINEN -- Saastamoinen model

MMF -- the Mean Mapping Function mode (Petrov (2009), in
preparation). Requires the name of the file with
the MMF model in the following qualifier.

2.39 WET_ZENITH_DELAY:

WET_ZENITH_DELAY: [NONE or zenith_delay_path_model_name]

Keyword WET_ZENITH_DELAY specifies the name of the a priori model for the non-hydrostatic component of the zenith path delay.

zenith_delay_path_model_name -- the name of the model. One of
NONE -- do not model non-hydrostatic path delay
MMF -- the Mean Mapping Function mode (Petrov (2009), in preparation). Requires the name of the file with the MMF model in the following qualifier.

2.40 HYDROSTATIC_MAPPING_FUNCTION:

HYDROSTATIC_MAPPING_FUNCTION: [NONE or mapping_function_name or [MMF mmf_mode file]]

Keyword HYDROSTATIC_MAPPING_FUNCTION defines the model for the mapping function for the hydrostatic component of the path delay -- dependence of the hydrostatic component of the atmospheric path delay with elevation. NB: the argument of the mapping function is so-called vacuum elevation: the elevation which the source would have in the absence of refractive atmosphere.

mapping_function_name -- the name of the mapping function. One of
NMFH -- Niell 1996 mapping function.
MMF -- the Mean Mapping Function mode (Petrov (2009), in preparation). Requires the name of the file with the MMF model in the following qualifier.

2.41 WET_MAPPING_FUNCTION:

WET_MAPPING_FUNCTION: [NONE or mapping_function_name or [MMF mmf_mode file]

or external_delay_dir]

Keyword WET_MAPPING_FUNCTION defines the model for the mapping function for the non-hydrostatic component of the path delay -- dependence of the non-hydrostatic component of the atmospheric path delay with elevation. NB: the argument of the mapping function is the so-called vacuum elevation: the elevation which the source would have in the absence of refractive atmosphere.

mapping_function_name -- the name of the mapping function. One of
NMF -- Niell 1996 mapping function.
MMF -- the Mean Mapping Function mode (Petrov (2009), in preparation). Requires the name of the file with the MMF model in the following qualifier.

2.42 ATMOSPHERE_TILT_PARTIALS:

ATMOSPHERE_TILT_PARTIALS: tilt_partials

Keyword ATMOSPHERE_TILT_PARTIALS defines the model for partial derivatives of the atmospheric tilt, or using another words, atmospheric gradients.

tilt_partials -- the name of tilt_partials model.
Currently, only one value is supported:
MACMILLAN_1995

2.43 SLANT_PATH_DELAY_MODEL

SLANT_PATH_DELAY_MODEL [SPD_3D or NONE]

Keyword SLANT_PATH_DELAY defines that the path delay through the neutral atmosphere for a given elevation, azimuth, time for a given station should be computed by three-dimensional B-spline interpolation

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of the time series of slant path delays. These time series of path delays are supposed to have been computed at an elevation/azimuth grid for each station using a standalone program SPD_3D by solving numerically a system of non-linear differential equations of wave propagation through the 4D, heterogeneous atmosphere (L. Petrov, 2009, in preparation) represented by the 4D field of atmospheric parameters that are the output of a numerical weather model. NB: the contribution of the neutral atmosphere can be computed either by using mapping function approach (keywords HYDROSTATIC_ZENITH_DELAY, HYDROSTATIC_MAPPING_FUNCTION, WET_MAPPING_FUNCTION, WET_ZENITH_DELAY) or by SLANT_PATH_DELAY.

SPD_3D -- slant path delay will be computed by interpolation of time series of B-spline coefficients stored in files that are kept directory EXTERNAL_DELAY_DIR (or EXTERNAL_DELAY_DIR_2, EXTERNAL_DELAY_DIR_3, EXTERNAL_DELAY_DIR_4 if defined)

NONE -- algorithm for computing slant path delay by interpolation of time series of B-spline coefficients will be disabled.

In a case if slant path delay is not available for one or more stations, the apriori path delay is computed in accordance with values of keywords HYDROSTATIC_ZENITH_DELAY and HYDROSTATIC_MAPPING_FUNCTION.

2.44 SLANT_PATH_DELAY_BIAS_FILE

SLANT_PATH_DELAY_BIAS_FILE [File or NONE]

File with a constant bias in the slant path delay.

2.45 ATMOSPHERE_PATH_DELAY_PARTIAL

ATMOSPHERE_PATH_DELAY_PARTIAL [NONE or NMW or NMF or IONO_350 or IONO_400 or IONO_450 or IONO_500 or TOTAL_SCALE or TOTAL_WATER or (GAUSSIAN_LAYER [layer_height layer_fwhm])]

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Keyword `ATMOSPHERE_PATH_DELAY_PARTIAL` defines partial derivative of slant path delay with respect to the path delay in zenith direction. This partial derivatives is not used for computing a priori path delay, but is supposed to be used for parameter estimation. The partial derivatives are placed in arrays `DER_DEL`, `DER_RAT`, in slots `VTD__DER_AT1` (partial derivatives for station #1) and `VTD__DER_AT2` (partial derivatives for station #2).

`NONE` -- no partial is computed

`NMFH` -- Niell (1996) hydrostatic mapping function

`NMFW` -- Niell (1996) wet mapping function

`IONO_350` -- mapping function for the case when residual atmosphere is described as a thin shell at height 350 km. Such a model is suitable for characterizing residual ionosphere.

`IONO_400` -- mapping function for the case when residual atmosphere is described as a thin shell at height 400 km. Such a model is suitable for characterizing residual ionosphere.

`IONO_450` -- mapping function for the case when residual atmosphere is described as a thin shell at height 450 km. Such a model is suitable for characterizing residual ionosphere.

`IONO_500` -- mapping function for the case when residual atmosphere is described as a thin shell at height 500 km. Such a model is suitable for characterizing residual ionosphere.

`TOTAL_SCALE` -- mapping function for the case when residual atmosphere is considered proportional to the total atmosphere. The partial derivative is defined as a ratio of the total slant path delay to the total path delay in the zenith direction. NB: this partial derivative requires

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computation of a priori slant path delay
(SLANT_PATH SPD_3D).

WATER_SCALE -- mapping function for the case when residual atmosphere is considered proportional to the water vapor contribution to the atmosphere. The partial derivative is defined as a ratio of the water vapor contribution of slant path delay to the water vapor contribution to path delay in the zenith direction. NB: this partial derivative requires computation of a priori slant path delay (SLANT_PATH SPD_3D).

GAUSSIAN_LAYER -- layer_height layer_fwhm -- mapping function for the case when the dependence of concentration of the residual atmosphere is described with the Gaussian model with the specified height and the specified full width half maximum (FWHM). The height of the layer in meters is specified in the 3rd qualifier and the FWHM in meter is specified in the 4th qualifier.

2.46 EXTERNAL_DELAY_DIR

EXTERNAL_DELAY_DIR [NONE or directory_name]

Keyword EXTERNAL_DELAY_DIR specifies the directory name of files with external path delays. The format of these files is defined by SLANT_PATH_DELAY keyword. If SLANT_PATH_DELAY is SPD_3D, then the files in the EXTERNAL_DELAY_DIR directory should satisfy spd_3d_bin specification. Each file contains the time series of the B-spline coefficients for a specific station.

2.47 EXTERNAL_DELAY_DIR_2ND

EXTERNAL_DELAY_DIR_2ND [NONE or directory_name]

Keyword EXTERNAL_PATH_DELAY_FILE specifies the secondary directory name of files with external path delays. Each file contains the time series of B-spline coefficients for a specific station. If a file for a given station is not found in directory EXTERNAL_DELAY_DIR, it is sought in directory EXTERNAL_DELAY_DIR_2ND.

2.48 EXTERNAL_DELAY_DIR_3RD

EXTERNAL_DELAY_DIR_3RD [NONE or directory_name]

Keyword EXTERNAL_PATH_DELAY_FILE specifies the name of third directory of files with external path delays. Each file contains the time series of B-spline coefficients for a specific station. If a file for a given station is not found in directories EXTERNAL_DELAY_DIR, EXTERNAL_DELAY_DIR_2ND, it is sought in directory EXTERNAL_DELAY_DIR_3RD.

2.49 EXTERNAL_DELAY_DIR_4TH

EXTERNAL_DELAY_DIR_4TG [NONE or directory_name]

Keyword EXTERNAL_PATH_DELAY_FILE specifies the name of fourth directory of files with external path delays. Each file contains the time series of B-spline coefficients for a specific station. If a file for a given station is not found in directories EXTERNAL_DELAY_DIR, EXTERNAL_DELAY_DIR_2ND, and EXTERNAL_DELAY_DIR_3RD, it is sought in directory EXTERNAL_DELAY_DIR_4TH.

2.50 IONOSPHERE_MODEL:

IONOSPHERE_MODEL: [NONE or GNSS_TEC_MAP]

Keyword IONOSPHERE_MODEL specifies the ionospheric model for computation of path delay. Accepted values:

NONE -- no ionosphere path delay is applied.

GNSS_TEC_MAP -- apply ionospheric path delay computed using GNSS TEC maps.

2.51 IONOSPHERE_SCALE:

IONOSPHERE_SCALE: value

Keyword IONOSPHERE_SCALE specifies the scale factor. The ionospheric contribution computed from TEC maps is multiplied by that parameter.

2.52 SOU_DEBIAS_MODEL:

SOU_DEBIAS_MODEL: [None value]

Keyword SOU_DEBIAS_MODEL specifies the name of the source position de-bias model. It is for compensation of the source position bias caused by the the deficiency of an ionospheric model or other models. The contribution of the de-bias model to path delay is computed on the basis of an empirical model that describes the position bias in declination as a function of declination. Supported models: NONE, MAP_850N, MAP_750R, MAP_780S.

2.53 IONOSPHERE_DATA_FILE:

IONOSPHERE_DATA_FILE: [NONE or file name]

Keyword IONOSPHERE_DATA_FILE specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.54 IONOSPHERE_DATA_FILE_2ND:

IONOSPHERE_DATA_FILE_2ND: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_2ND specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.55 IONOSPHERE_DATA_FILE_3RD:

IONOSPHERE_DATA_FILE_3RD: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_3RD specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.56 IONOSPHERE_DATA_FILE_4TH:

IONOSPHERE_DATA_FILE_4TH: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_4TH specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.57 IONOSPHERE_DATA_FILE_5TH:

IONOSPHERE_DATA_FILE_5TH: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_5TH specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.58 IONOSPHERE_DATA_FILE_6TH:

IONOSPHERE_DATA_FILE_6TH: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_7TH specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.59 IONOSPHERE_DATA_FILE_8TH:

IONOSPHERE_DATA_FILE_8TH: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_8TH specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.60 IONOSPHERE_DATA_FILE_9TH:

IONOSPHERE_DATA_FILE_9TH: [NONE or file_name]

Keyword IONOSPHERE_DATA_FILE_9TH specifies a binary file ion viono format that keeps GNSS TEC maps for some period of time.

2.61 GRS_METRIC:

GRS_METRIC: metric_identifier

Keyword GRS_METRIC specifies the name of the metric tensor for the geocentric reference system.

metric_identifier -- Identifier of the metric tensor. Supported values:

IAU2000 -- the metric tensor proposed by Kopeikin and Brumberg. recommended by IAU2000 resolution.

ITRF2000 -- the metric tensor which was implicitly used in data analysis in the 20-th century, which was used for constructing the ITRF2000 catalogue of station positions and velocities.

Comment: recommendation of IERS2003 is contradictory: in one place the authors of Conventions claimed that they compliant with the IAU2000 recommendation, at another place they propose formulae based on ITRF2000 metric.

2.62 TROP_AXOF_COUPLING:

TROP_AXOF_COUPLING: usage_flag

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The keyword TROP_AXOF_COUPLING specifies whether to compute contribution to delay due to coupling of the troposphere path delay and antenna axis offsets. By another words TROP_AXOF_COUPLING takes into account additional troposphere path delay due to motion of the reference point.

usage_flag -- the flag. Supported values:

- YES -- to compute contribution due to coupling of the troposphere path delay and antenna axis offsets.
- NO -- not to compute contribution due to this coupling.

2.63 TROP_GEOMETRIC_COUPLING:

TROP_GEOMETRIC_COUPLING: usage_flag

The keyword TROP_GEOMETRIC_COUPLING specifies whether to compute the contribution to delay due to coupling of the troposphere path delay and geometric delay.

usage_flag -- the flag. Supported values:

- YES -- to compute contribution due to coupling of the troposphere path delay and geometric delay.
- NO -- not to compute contribution due to this coupling.

2.64 PARALLACTIC_ANGLE:

PARALLACTIC_ANGLE: usage_flag

The keyword PARALLACTIC_ANGLE specifies whether to compute the contribution to phase delay and phase delay rate due to the parallactic angle. NB: the contribution of the parallactic angle to group delay is zero.

usage_flag -- the flag. Supported values:

- YES -- to compute contribution to phase delay due

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parallactic angle
NO -- not to compute contribution due to parallactic angle.

2.65 GALACTIC_ABERRATION:

GALACTIC_ABERRATION: *usage_flag*

The keyword GALACTIC_ABERRATION specifies whether to compute the contribution to delay caused by the galactic aberration.

usage_flag -- the flag. Supported values:
YES -- to compute contribution to phase delay due
parallactic angle
NO -- not to compute contribution due to parallactic angle.

2.66 GEOM_EXPR_FAR_ZONE:

GEOM_EXPR_FAR_ZONE: *model_name*

The keyword GEOM_EXPR_FAR_ZONE specifies the name of the expression for geometric delay in far zone (larger than 10 pc) in vacuum based on general relativity.

model_name -- the name of the expression. Supported values:
KS_1999 -- expression Kopeikin and Schaefer, 1999
PK_2001 -- expression Petrov and Kopeikin, 2001 (recommended)

2.67 GEOM_EXPR_NEAR_ZONE:

GEOM_EXPR_NEAR_ZONE: *model_name*

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The keyword `GEOM_EXPR_NEAR_ZONE` specifies the name of the expression for geometric delay in the near zone (from Earth's satellite to 10 pc) in vacuum based on general relativity.

`model_name` -- the name of the expression. Supported values:

`LIGHT_TIME` -- iterative solution of the light equation

`JN_2019` -- Jaron, Nothnageeel expersion

2.68 SOURCE_STRUCTURE:

SOURCE_STRUCTURE: `control_file_name`

The keyword `SOURCE_STRUCTURE` specifies the control file for modeling contribution to delay due to source structure.

`control_file_name` -- the name of the control file or

`NO` -- contribution to delay due to source structure
will not be computed.

This option is currently being tested.

2.69 DOPPLER_EXPR:

DOPPLER_EXPR: `model_name`

The keyword `DOPPLER_EXPR` specifies the name of the expression for computation of the Doppler frequency shift for an object within the Solar system based on general relativity.

2.70 DELAY_RATE:

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DELAY_RATE: YES or NONE

The keyword DELAY_RATE specifies whether to compute delay rate.
If DELAY_RATE is NONE computation runs faster.

YES -- to compute delay rate using analytical expressions.

NO -- not to compute delay rate.

Questions and comments about this guide should be directed to:

Leonid Petrov (<http://astrogeo.org/petrov>)

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